Ex: 1	Private Cloud using OpenStack
Date:	

Aim

This experiment focuses on setting up and configuring a private cloud infrastructure by deploying a single-node OpenStack instance using MicroStack. The objective is to gain practical knowledge of fundamental cloud services, including compute, networking, and storage resources, by launching and interacting with virtual machine instances within this environment.

What is OpenStack?

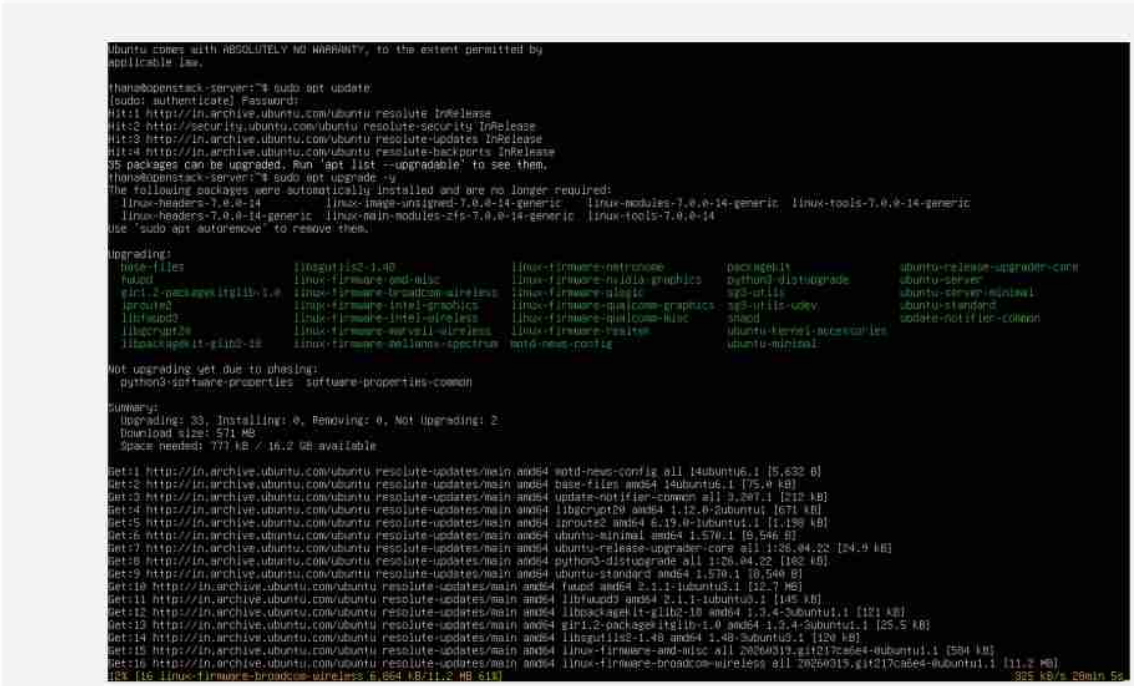
- OpenStack functions as an open-source, free standard for cloud computing.
- It is predominantly utilized to provide Infrastructure-as-a-Service (IaaS) solutions.
- This platform facilitates the on-demand provisioning and management of virtual machines (VMs).

What is MicroStack?

MicroStack provides a streamlined, single-node local execution of OpenStack. It serves as an excellent tool for educational purposes, experimental testing, or small-scale development. The platform simplifies the deployment and validation of OpenStack services on hardware ranging from laptops to compact servers.

Procedure

Step 1: Establish an Ubuntu VM within Oracle VirtualBox using these specifications: 8 GB RAM, 4 CPU Cores, and 30 GB of disk space.
Step 2: Complete the Ubuntu OS installation, then sign in with your configured credentials. ![Ubuntu Installation]
Step 3: Refresh the local package repository and upgrade existing software using the commands: 'sudo apt update' followed by 'sudo apt upgrade'.



Step 4: Deploy the MicroStack environment via the beta channel using the command: 'sudo snap install microstack --beta'.

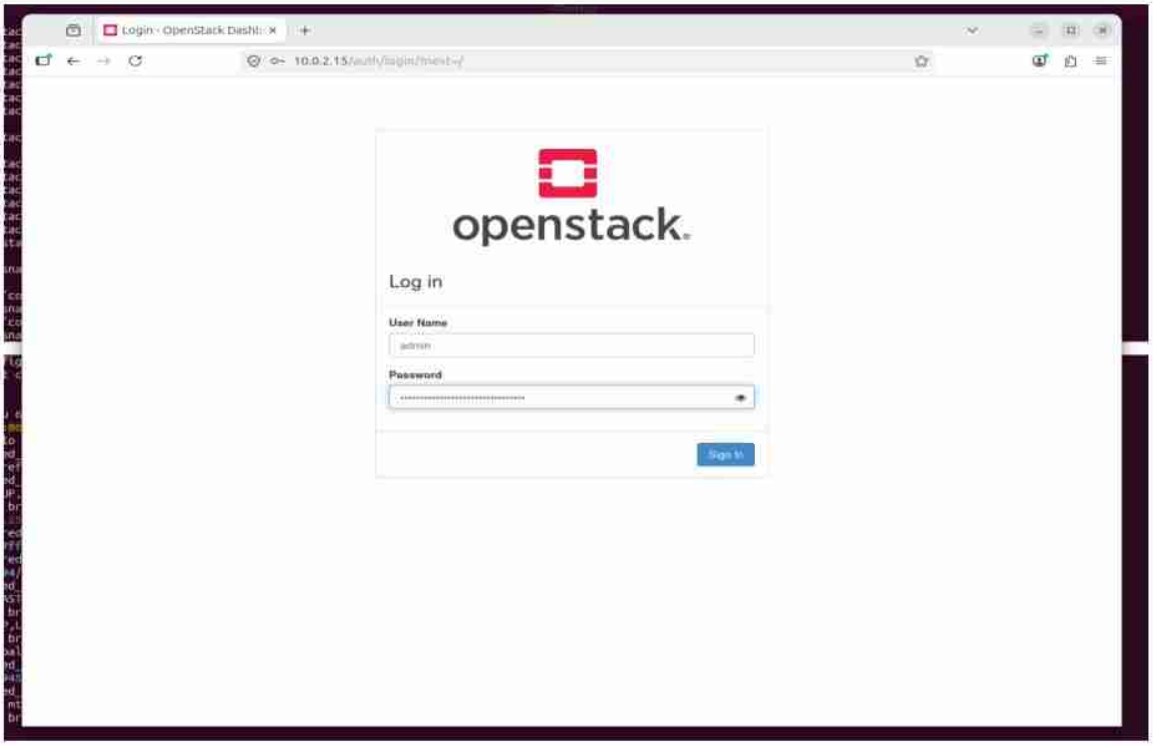
Step 5: Confirm the MicroStack installation status with: 'snap list microstack'.

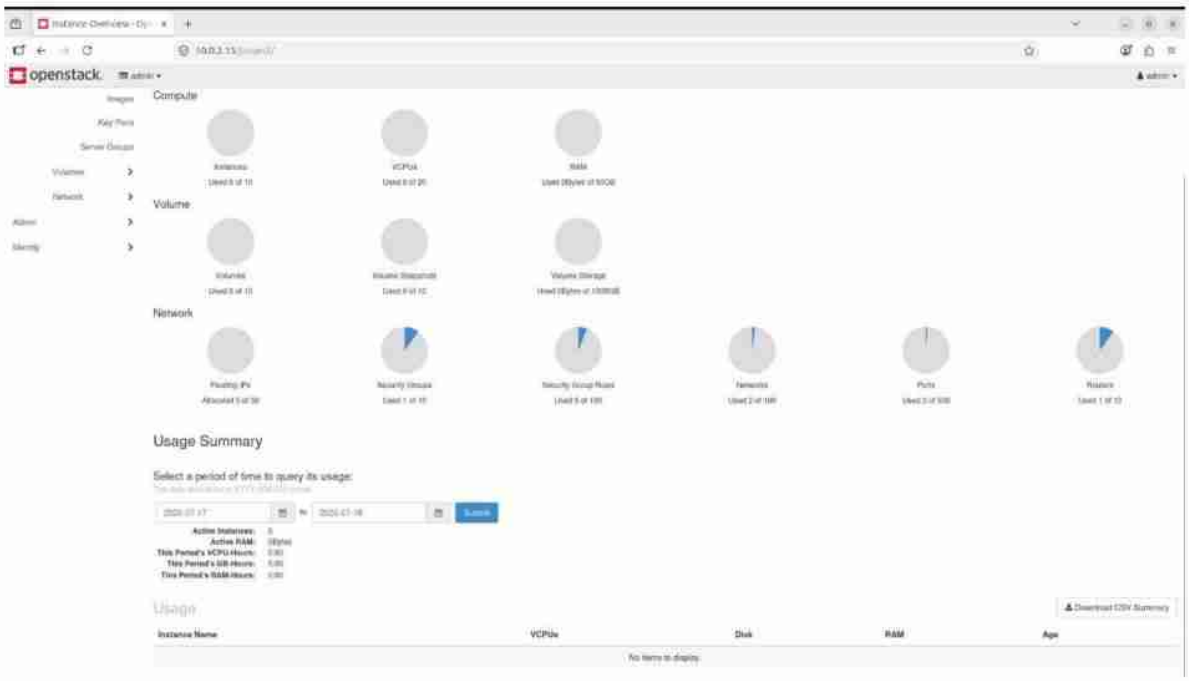
Step 6: Execute the initialization process for OpenStack services by running: 'sudo microstack init --auto --control'.

Step 7: Check the currently installed version of OpenStack: 'microstack.openstack --version'.

Step 8: Fetch the administrative Keystone password with this command: 'sudo snap get microstack config.credentials.keystone-password'.

Step 9: Access the OpenStack Horizon Dashboard via your web browser by navigating to the designated IP address displayed post-initialization. Use 'admin' as the username and the password retrieved in the previous step to authenticate.





Conclusion

The installation and configuration of MicroStack on Ubuntu were completed successfully, confirming that the Horizon Dashboard is fully operational.